Prajwal Nayak | **22B4246**
Electrical Engineering | **Dual Degree (B.Tech. + M.Tech.)**
Indian Institute of Technology Bombay | **Gender: Male**
Specialization: Communication & Signal Processing | **DOB: 25/06/2004**

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Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2027	8.59 Core: 204 | Total: 309

- Completed a Minor in **Artificial Intelligence and Data Science** from **C-MinDS, IIT Bombay** *('25)*

RESEARCH INTERESTS

Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Optimization in Machine Learning, Game Theory, Anomaly Detection, Audio Signal Processing, Speech Processing, Digital Signal Processing, Time Series Analysis and Forecasting, Data Analytics, Energy Systems Optimization, Smart Grid Technologies, Renewable Energy Analytics, Data Clustering and Dimensionality Reduction, Probabilistic Machine Learning, Statistical Machine Learning, Self-Supervised Learning, Biomedical Signal Processing, Audio Synthesis and Coding, Energy Informatics, Smart Grid Analysis, Theoretical Machine Learning

SCHOLASTIC ACHIEVEMENTS

- Secured a **99.47 percentile** in JEE Main, demonstrating exceptional proficiency among a million candidates *('22)*
- Achieved All India Rank 4691, translating to a **96.98 percentile** in the highly rigorous JEE Advanced exam *('22)*
- Attained state rank 219 in KCET, achieving a **99.89 percentile** and demonstrating true academic excellence *('22)*
- Awarded **Branch Change** (**Top 7%**) in freshmen year owing to exemplary overall academic performance *('23)*
- Consistently achieved **AA**/**AB** grades in **30+** multidisciplinary courses across 8 rigorous academic semesters *('26)*
- Earned the **Gold Medal**, a symbol of academic & extracurricular merit for **10 consecutive years** at school *('19)*

CONFERENCES AND RESEARCH PROJECTS

Energy Informatics Academy (EI.A) | ***Research Paper*** *Jul'26*
Accepted for presentation at EI.A 2026, Beijing and publication in Springer Proceedings

- Optimized energy storage for a **renewable energy community** of 1,833 households and an **IITB Data Center**
- Developed a **storage sizing and energy-sharing framework** across four centralized/decentralized configurations
- Demonstrated how **storage topology and battery costs** influence grid dependence and local energy utilization

Ultra-Low-Bitrate Neural Audio Codec | ***Dual Degree Project*** *Mar '26 – Present*
Prof. Srikanth Chetupalli, Department of Electrical Engineering, IIT Bombay

- Implemented **LSPnet**, a 1.2 kbps neural speech codec using deep learning for high-fidelity, low-complexity coding
- Developed a **Continuous Flow Matching (CFM)** decoder to improve quality in neural generative audio models
- Implementing **stochastic interpolants** with latent-space representations for efficient probability-flow sampling, compressed-state modeling, and **very-low-latency inference**, targeting robust **ultra-low-bitrate** communication

High-Performance Computing Cluster | ***Internship at THFT Laboratory*** *May '24 – Jul '24*
Prof. Janani Srree Muralidharan, Department of Mechanical Engineering, IIT Bombay

- Configured a **SLURM HPC cluster** on Debian 12 with one master and two compute nodes for parallel workloads
- Integrated **NFS shared storage** & **Munge authentication** for synchronized data access & secure job execution
- Implemented **static networking** and **virtual bridging** to enable isolated communication across compute nodes

COURSE PROJECTS

AI Guard Agent "Sentinel" | ***Advanced Machine Learning, EE782*** *Oct '25*

- Implemented a real-time **AI room guard** using deep learning for authorized personnel identification and monitoring
- Developed a **Python face recognition pipeline** for detection and encoding under varied environmental conditions
- Integrated **anomaly detection** and real-time alerts to identify unusual activities and strengthen system security
- Evaluated **precision, recall, and inference latency** to optimize recognition performance & system responsiveness

GeoVision: Remote Sensing Image Captioning | ***Advanced Machine Learning, EE782*** *Sept '25*

- Built **CNN-LSTM and CNN-Transformer** captioning models using pretrained ResNet-18 and MobileNet backbones
- Developed **caption tokenization** & **sequence-generation pipelines** for end-to-end model training and evaluation
- Tuned **embeddings, hidden dimensions, and dropout** to optimize model performance on the RSICD dataset
- Evaluated models using **BLEU-4**, with Transformer architectures achieving better performance over LSTM variants

WavePrint: Contact-Based Fingerprint Feature Learning | ***Prof. Vikram Gadre*** *Oct'24– Nov'24*

- Integrated **Haar wavelet transforms** with **encoder-decoder convolutional neural network architecture** for **multi-scale** fingerprint feature extraction, denoising, fine-detail preservation, and efficient representation learning
- Developed **feature embeddings** combining texture & minutiae information from contact-based fingerprint images
- Explored **contrastive learning** using MoCoV2 & AIRMeC for robust & discriminative fingerprint feature extraction

NeuroDSP: Interpretable DSP-CNN for ECG Analysis | ***Prof. Vikram Gadre*** *Oct '25 – Nov '25*

- Devised a **DSP-CNN architecture** using learnable bandpass filters for ECG classification from time-series signals
- Implemented **attribution mapping** to quantify feature relevance and identify influential segments of ECG signals
- Parameterized convolutional bandpass filters with **learnable cut-off frequencies** enabling spectral regularization

eMARC: Energy Consumption & Household Personas | ***Energy Data Analytics, EE467*** *Sept '25*

- Performed **exploratory data analysis** & data wrangling on given time-series household energy consumption data
- Engineered normalized features including Energy Use Intensity, Per-Capita EUI, and Seasonal Ratio for clustering
- Applied **Gaussian Mixture Models (GMM)** and identified three distinct household energy consumption personas
- Utilized **Dynamic Time Warping** with hierarchical clustering to group households by consumption shape similarity

SolarCast: Solar Power Generation Forecasting | ***Energy Data Analytics, EE467*** *Oct '25*

- Analyzed **time-series solar data** using ADF, KPSS, & spectral analysis to identify daily, & yearly seasonal patterns
- Developed **SARIMA baseline** with log, quantile, & Yeo-Johnson transformations for skewed, zero-inflated features
- Built a **two-stage ML pipeline** using Random Forest and XGBoost, achieving over 99.8% classification accuracy
- Benchmarked the **ML models against SARIMA**, demonstrating superior forecasting accuracy for solar generation

Modular Digital Logic Equivalence Checker | ***VLSI CAD, EE677*** *Dec '25*

- Engineered a hybrid **Equivalence Checker** combining Python with a C++ backend for efficient logic simulation
- Automated **Miter Circuit generation** from VHDL and Bench netlists, implementing ID shifting to prevent collisions
- Designed a **heuristic based** solver using back-tracing and DFS algorithms to isolate counter-example test vectors
- Integrated an **Espresso logic minimizer**, utilizing expand, irredundant, & reduce to optimize SOP expressions

BootSAR: 10-bit 200 MS/s Differential SAR ADC | ***Prof. Maryam Shojaei Baghini*** *Feb'26 - Mar'26*

- Designed differential bootstrapped switch, 45nm CMOS, achieving 11.82-bit ENOB and 72.97 dB SNR at 200MS/s
- Sized **hold capacitors and split-capacitor CDAC** using thermal-noise and settling constraints for 10-bit accuracy
- Implemented **asynchronous SAR logic and CRS switching**, reducing average DAC switching energy by 15
- Integrated **SAR ADC** and characterized performance using FFT, achieving 62.475 dB SNDR and 10.08-bit ENOB

MIRAGE: Motion-Integrated Articulated Glove Emulator | ***Prof. Siddharth Tallur*** *Jan '25 – Apr '25*

- Designed a **1°precision 3D-printed exoskeleton glove** using rotary encoders and Rpi Pico W for finger tracking
- Implemented **real-time UDP wireless transmission** in Python for low-latency motion streaming and simulation
- Developed a **Unity 3D hand model** using Blender rigging and C# scripting for accurate real-time hand articulation
- Designed a **custom PCB with multiplexers** using KiCad, gaining experience in circuit design and PCB fabrication

FPGA-DSP: Open-Source DSP Teaching Kit | ***Applications of DSP, EE603*** *Sept '25 – Nov '25*

- Investigated **FPGA implementations of FIR, FFT, and CORDIC** using open-source hardware design toolchains
- Developed **parameterizable streaming DSP modules** in Verilog with automated testbenches for Icarus Verilog
- Built **Q-format fixed-point analysis scripts** to simulate quantization noise and meet target signal-to-noise ratios

RTL-OFDM: 16-QAM Modulator & Transmitter | ***Prof. Laxmeesha Somappa*** *Nov '25*

- Built a synthesizable **16-QAM module in Verilog** using Q5.10 fixed-point arithmetic for hardware implementation
- Constructed a **subcarrier mapping unit** to organize the six QAM symbols into a 16-point IFFT input sequence
- Implemented a **radix-2 DIT 16-point IFFT module** in Verilog for frequency-to-time domain OFDM signal generation
- Integrated the **OFDM transmitter pipeline** and validated functionality using testbenches and GTKWave waveforms

SwitchCraft: Arbiter-Based 2×2 Packet Switch | ***Algorithmic Design of Digital Systems, EE789*** *Feb '26*

- Developed three **2×2 switch architectures** using queues, arbiter muxes, and demuxes for distinct routing paths
- Implemented **priority arbitration** with packet-level priority toggling to ensure access when both inputs are active
- Validated all switch designs using **AHIR testbenches and GHDL simulation**, confirming reliable packet transfers

IITB-CPU: 16-bit RISC Processor | ***Microprocessors, EE309*** *Jan 2024*

- Designed a **16-bit RISC processor** featuring a 6-stage pipeline supporting R-Type, I-Type, and J-Type instructions
- Implemented data forwarding paths & stall mechanisms to mitigate **data hazards** & optimize pipeline performance
- Developed datapath and control logic, integrating 8 general-purpose registers (with R0 acting as Program Counter)

POSITIONS OF RESPONSIBILITY

Saathi, IIT Bombay | ***Overall Head*** | ***Convener*** *Present* | *Jul'26 - Jul'27*

Queer support and resource group at IIT Bombay

- Spearheading a **7-member core team and 60+ volunteers/coordinators** with an overall budget over INR 2.5L
- Organized **Rangavali**, IIT Bombay's annual queer fest, successfully executing 10+ events with 1,000+ attendees
- Drove **90% capacity attendance** through collaboration with the Mumbai International Queer Film Fest initiative

Teaching Assistant | ***Communication Systems, EE341*** *Aug'26 – Present*

- Managed **course logistics for 100+ students**, coordinating evaluations, attendance & academic communications
- Supported **course instruction and assessment**, assisting students with concepts, and resolving their queries

Institute Technical Convener | ***Tinkerers' Lab, IIT Bombay*** *Apr '23 – Apr '24*

- Managed **INR 60L+ equipment inventory** and lab operations, overseeing procurement, maintenance, and access
- Orchestrated technical workshops on **SolidWorks, LaserCAD, Conda, Python and PyTorch** for **100+ students**

Project Mentor, Seasons of Code (SoC) | ***Web and Coding Club, IIT Bombay*** *May '26 – Jul '26*

- Mentored student teams in developing an **advanced AI agent for security** as part of the SoC 2026 joint initiative
- Guided mentees through agile development, providing support with **technical implementation and debugging**

Project Mentor, XLR8 | ***Electronics & Robotics Club, IIT Bombay*** *Sept '24*

- **Mentored freshman teams** in designing manually controlled robots for IITB's flagship robot making competition
- Guided mentees through **hardware integration & circuit design**, providing technical training with **Arduino Nano**
- Assisted teams in optimizing mechanical constraints and control systems to navigate complex obstacle courses

RELEVANT COURSEWORK

ML and CS	Optimization in ML, Advanced Topics in Machine Learning, Introduction to Machine Learning, Programming for Data Science, Game Theory & Algorithmic Mechanism Design
Signal Processing	Speech Processing, Advanced Image Processing, Fundamentals of Digital Image Processing, Digital Signal Processing & its Applications, Wavelets, Signal Processing I
VLSI & Architecture	Mixed-Signal VLSI Design, VLSI Design, Foundation of VLSI CAD, Algorithmic Design of Digital Systems, Processor Design, Microprocessors, Digital Systems
Communications	Information Theory & Coding, Communication Systems I & II, Wireless & Mobile Communication, Radar Systems
Power & Energy	Switchgear Principles, Energy Data Analytics, Power Engineering I, Power Engineering II
Math	Probability and Random Processes, Linear Algebra, Differential Equations, Calculus I & II

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, Verilog, VHDL, Assembly (8051), HTML/CSS
AI, DS and ML	PyTorch, TensorFlow, Scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Seaborn, NLTK, OpenCV
DSP / Hardware	Cadence Genus, Icarus Verilog, GTKWave, Magic, NGSPICE, LTSpice, Quartus, GNU Radio
Tools / Infra	AWS, Git, PyQt, SLURM, NFS, Munge, Fusion 360, SolidWorks, μVision

EXTRACURRICULAR ACITIVITIES

- Volunteered for the **NSS Cloth Collection Drive** 2022, coordinating door-to-door collection for donation distribution
- **Linguist:** understanding of **5 languages**: English, Kannada, Hindi, Konkani, Tulu; learning **Japanese & Spanish**
- Attended the **ENERGIZE International Conference 2025**, engaging with research in energy and sustainability